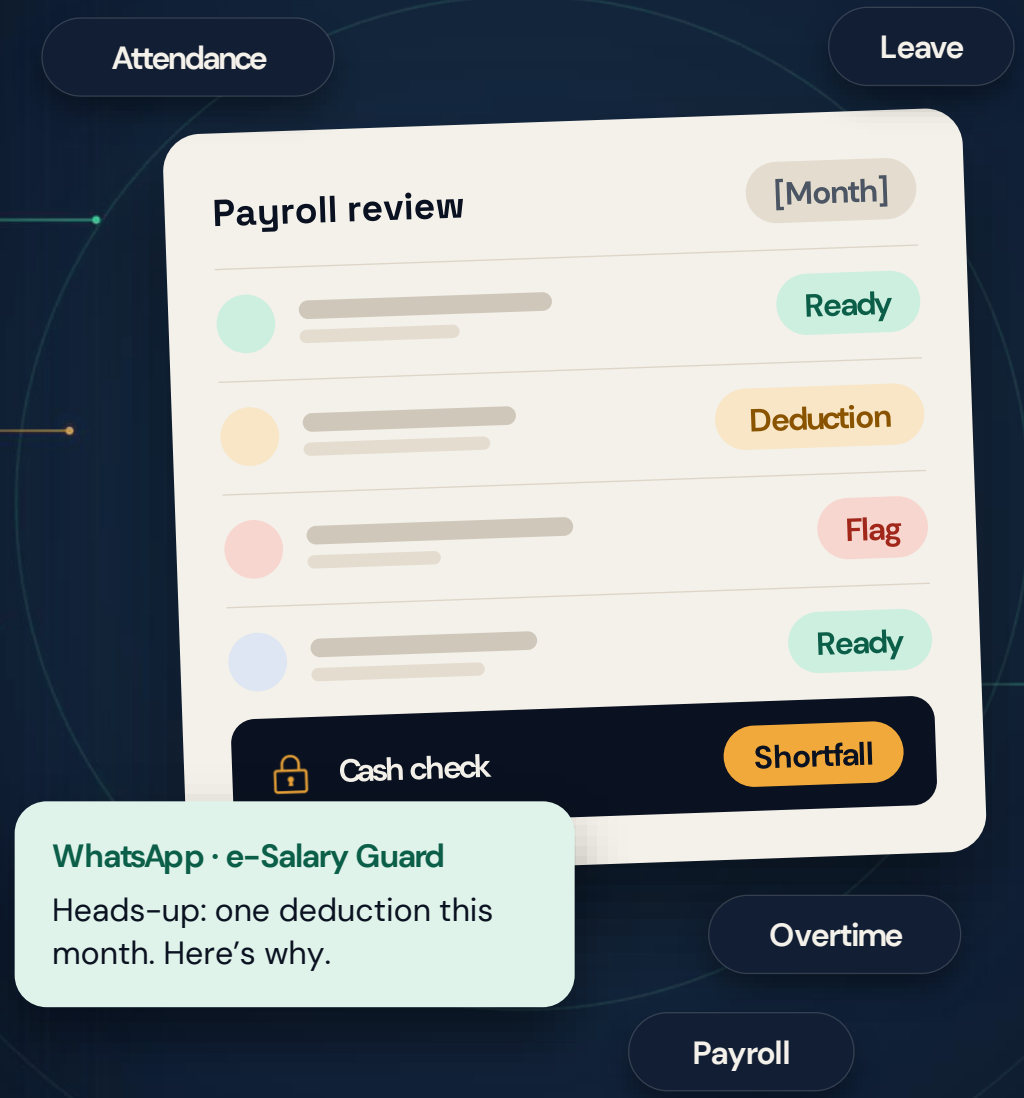
SDC HACKATHON 2026

e-Salary Guard

Payroll calculated right, explained early, and funded before payday.



Team SPNG · SDC Hackathon 2026

02 — THE PROBLEM

Payroll data lives in separate places

Attendance ----- Leave ----- Overtime ----- Payroll

HR

Calculates by hand

Attendance, leave and overtime sit in different systems, so every payroll run is manual and error-prone.

Employee

Find out on payday

Deductions only appear on the payslip, which turns payday into complaints.

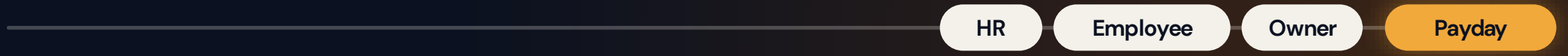
Owner

Finds out cash is short

The payroll total is known at the last minute, too late to plan cash.

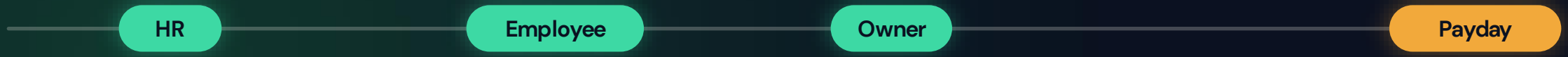
Every payday, three people find out too late: the HR officer, the employee and the owner.

TODAY



e-Salary Guard moves that moment earlier, for all three.

WITH E-SALARY GUARD



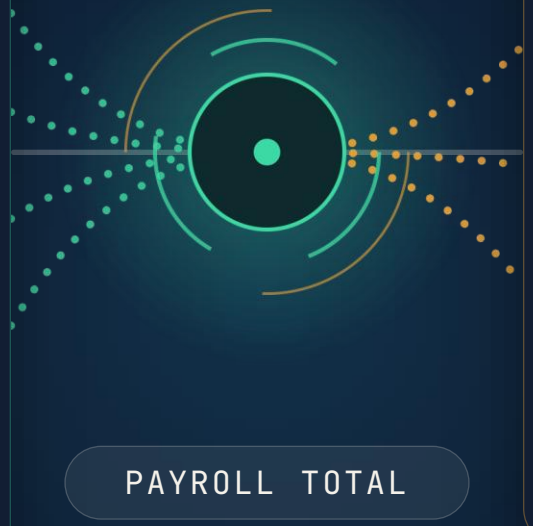
05 — THE SOLUTION

One engine, two guards

Salary Guard

FOR HR Employee

- Calculates pay from Orbyte attendance plus HR-approved leave and overtime
- Warns workers on WhatsApp as soon as a deduction happens
- Payroll review with flags, then one-click pay



Cash Guard

FOR Owner

- Compares the payroll total with the cash balance before payday
- Adds the owner's estimate of other payday costs
- Warns the owner early if cash will fall short

06 — WORKFLOW

How it works

1 Attendance

Check-in, check-out and lateness come from Orbyte.

2 HR-approved data

Only approved leave and overtime, tagged paid or unpaid.

3 HR control

HR sets base pay, bonus, overtime rate and deduction rules.

4 Calculation

Pay is recalculated whenever data or a rule changes.

5 WhatsApp warning

Workers with a deduction get a message saying why.

6 Payroll review

All pay in one table, with anomalies flagged.

7 Cash check

Cash Guard: total payroll vs. cash balance, before payday.

8 One-click pay

Approve, confirm the total, pay. Payslips via WhatsApp.

9 Activity log

Every change, calculation, payment and message is logged.

Simple rules, applied every time

PAY CALCULATION

+ Base salary

+ Approved overtime

+ Bonus

- Attendance deductions

Pay — gross, per worker

What counts as a deduction

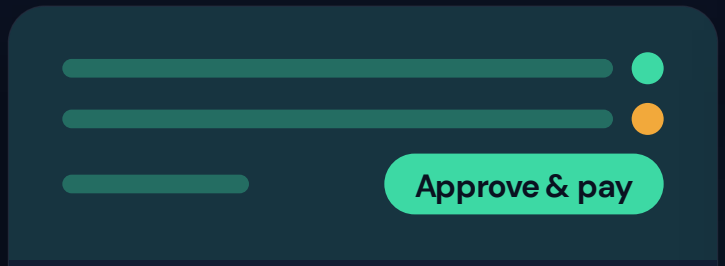
Late arrivals, absence without approved leave, and leave HR marked as unpaid.

What the system never does

Judge a leave reason. It only reads what HR approved, so the decision stays with people.

08 — WHO USES IT

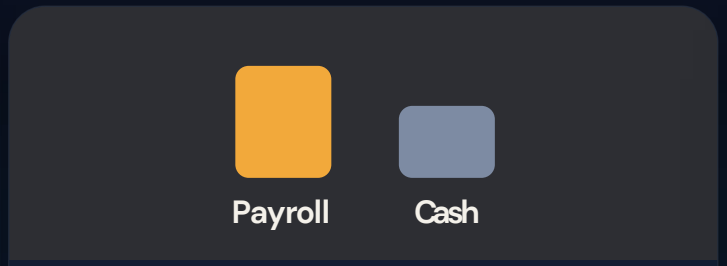
Three views, one database



WEB LOGIN

HR

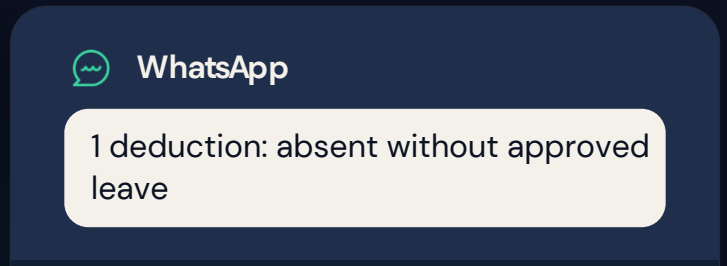
Sets pay rules, reviews payroll with flags, approves and pays.



WEB LOGIN

Owner

Sees payroll against cash, gets a shortfall warning and a monthly summary.



NO LOGIN · WHATSAPP

Employee

Gets deduction warnings with the reason, and the payslip on payday.

Where AI fits



Explains deductions

Turns a deduction into a plain-language WhatsApp message a worker understands.

Flags anomalies

Spots a sharp pay drop or unusual overtime in payroll review, before HR pays.

Briefs the owner

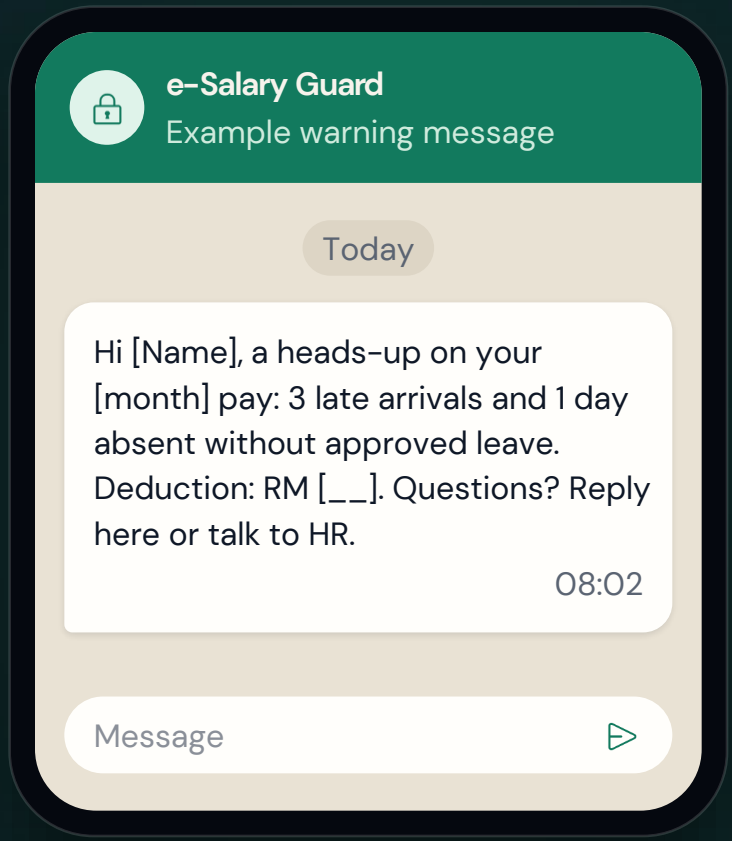
Summarises what changed this month and whether cash covers payday.

Rules decide the money; HR makes the final call. AI never approves leave or changes a number.

10 — LIVE DEMO

One payday, start to finish

1. Synthetic attendance loads for a small team.
2. One worker is late 3 times and absent 1 day without approval.
3. The deduction is calculated automatically.
4. **The worker gets a WhatsApp warning with the reason.**
5. HR opens payroll review and sees the flag.
6. **Cash check shows a shortfall; the owner is warned in time.**
7. HR confirms the total and pays (simulated).
8. Payslips go out on WhatsApp; the period locks.



Built on what SMEs already use

Reads Orbyte attendance. No new system, no re-entry.

Reusable modules. Pay calculation and WhatsApp alerts work for any SME.

HR approval stays the source of truth. We read decisions; we don't replace them.

</> How we build it

HOSTING	Replit + RunPod
CODING	AI-assisted, with Claude
DATA	Synthetic attendance, simulated approvals
PAYMENTS	Simulated in the MVP
WHATSAPP	Simulated in the MVP

12 — SCOPE AND ROADMAP

MVP scope and what comes next

AREA	In the MVP	Next
Pay calculation	Gross pay from attendance and approved data	Statutory deduction modules per country (in Malaysia: EPF, SOCSO, EIS, PCB)
Cash check	Payroll vs. balance, plus the owner's estimate of other payday costs	Employer contributions calculated automatically
Data	Synthetic attendance, simulated approvals	Live Orbyte and e-leave integration
Payments	Simulated, behind a confirmation screen	A real payment gateway
WhatsApp	Simulated messages	WhatsApp Business API

13 — IMPACT

What changes for an SME

Less manual HR work

Fewer pay errors

Workers know before payday

Owners know cash is ready

8 UN SUSTAINABLE DEVELOPMENT GOAL

Decent Work and Economic Growth

Fair, transparent pay is part of decent work: every deduction comes with its reason, before payday.

Five people, clear ownership

T1

Backend and data

Manuel Otto Chandra

T2

AI agent and integrations

Dixson Oswaldo Tanjaya

T3

Frontend and UX

Gregorius Marvel Wicaksana

C1

Commercial: Business Model

Owen Ferdinand Lee

C2

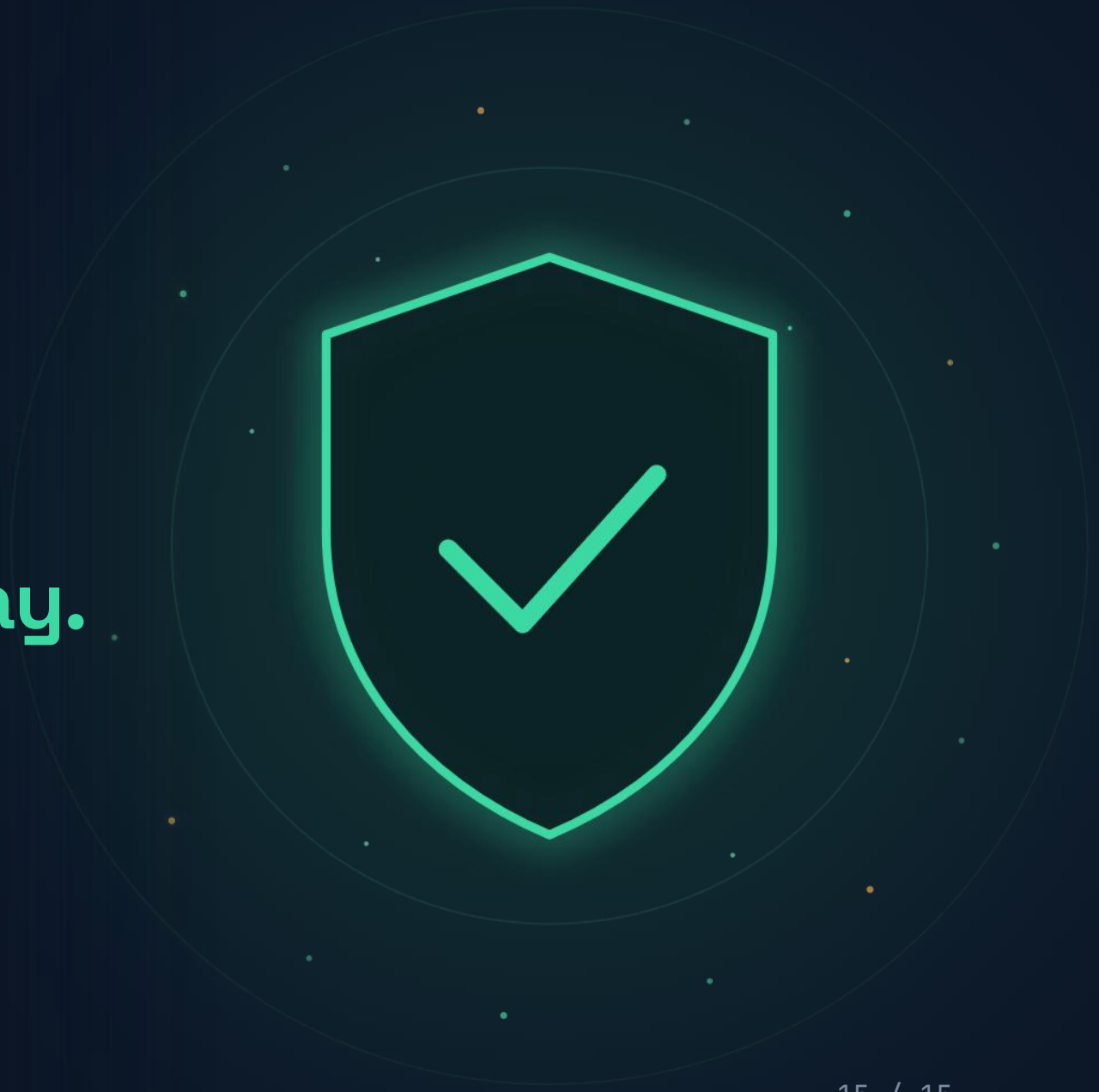
Commercial: Proposal and Document

Sunu Adi Sotya

Three on the product, two on the business.

E-SALARY GUARD

Pay calculated right.
Workers told early.
Cash ready before payday.



Thank you. Questions?